

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.2 THE ATOM: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Your driving question is: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? Write a clear explanation in the blank space under each prompt. Use evidence from your lessons: the flame tests, the gold foil simulation, your Bohr and orbital drawings, the isotope and relative atomic mass calculations, and your electron configurations. Refer to the Rift Valley cooking hearth and the Kisumu school laboratory where you can. Write in full sentences, use correct chemistry words, and show any working. Your teacher will mark your answer against the rubric.

Part 1: The Puzzle and What Atoms Are Made Of

Describe what you saw at the start of the unit when salt, banana-peel ash and different substances were put in a flame. Why did this seem puzzling when you knew that all substances are made of atoms and fire is just heat? Then use the Rutherford gold foil simulation to explain how scientists learned that the atom has a nucleus. For each of the three observations (most particles, a small number, a very small number), say what it told scientists about the atom. Then say what the Rutherford model could not yet explain about the flame colours.

At the Rift Valley hearth, spilled salt (sodium chloride) made the fire glow a brilliant yellow-orange, and banana-peel ash (rich in potassium) gave a soft violet flash. In the Kisumu laboratory we saw that sodium chloride gave a yellow-orange flame, potassium permanganate gave violet, and copper sulphate gave blue-green, and these colours were repeatable for each substance we tested. This was puzzling because all substances are made of atoms and fire only supplies heat, so it was not clear why the type of atom should change the colour. In the Rutherford gold foil simulation, most alpha particles passed straight through the foil with little or no deflection, so the atom is mostly empty space. A small number were deflected at moderate angles, which showed that there is a concentrated positive charge in the atom that repels the positive alpha particles. A very small number bounced back at large angles, which showed that this positive charge and most of the mass are packed into a tiny, dense nucleus at the centre, with electrons in the space around it. This forced scientists to revise Dalton's solid-sphere model. The atomic number is the number of protons, and the mass number minus the atomic number gives the number of neutrons. However, the Rutherford model only says that electrons are somewhere outside the nucleus. It could not explain where the electrons are or why different substances give different flame colours, so we needed a model of electron arrangement.

Part 2: Energy Levels and the Origin of Flame Colour

Using Bohr's model, explain step by step what happens to the outermost electron of a

Sodium has atomic number 11, so it has 11 electrons arranged 2, 8, 1. In Bohr's model electrons can only occupy fixed energy levels (shells), with a maximum of 2 electrons in shell 1 and 8 in shell 2, and they cannot exist between shells. Step 1: the flame supplies heat energy. Step 2: the outermost electron in shell 3 absorbs this energy and jumps to a higher energy level. This is the

sodium atom when salt falls into the cooking fire. Say how many electrons sodium has in each shell and write its electron arrangement. Explain why the light that comes out is a particular colour, and why potassium gives a different colour from sodium.	excited state. Step 3: the excited state is unstable, so the electron falls back down to a lower energy level. Step 4: the energy released in the fall is given out as a photon of light. The colour of the light depends on the exact amount of energy released, which depends on the gap between the two energy levels involved in the fall. For sodium, the fall that gives the strong yellow-orange line is from the excited level back to the level of the outermost electron, and its energy gap corresponds to light of about 589 nm. Potassium atoms have a different electron arrangement and differently spaced energy levels, so the gap, and therefore the photon energy, is different and the light we see is violet. Each element has its own pattern of energy level gaps, so each gives its own characteristic colour. Flame colour is therefore a fingerprint of the electron structure of the atom.
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Part 3: Orbitals and Electron Configuration

(a) Explain why the Bohr shell model was not enough and what s and p orbitals add. (b) State the three rules that decide how electrons fill orbitals. (c) Write the s/p electron configuration of sodium and of potassium, and explain how the configurations help to explain the different flame colours of sodium and of potassium.	(a) Bohr shells could not explain the energy differences within a shell. The orbital model says that within each principal energy level, electrons occupy sub-regions called orbitals. An s orbital is spherical and holds a maximum of 2 electrons. There are three p orbitals (px, py, pz), each holding 2 electrons, so 6 electrons in total in the p sub-level. (b) The Aufbau principle: electrons fill the lowest-energy orbital first (1s, 2s, 2p, 3s, 3p, and so on). The Pauli exclusion principle: each orbital holds at most 2 electrons and they must have opposite spins. Hund's rule: when orbitals have equal energy, such as the three 2p orbitals, one electron goes into each before any orbital gets a second electron. (c) Sodium (Z = 11): $1s^2 2s^2 2p^6 3s^1$. Potassium (Z = 19): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. Both have one lone outermost s electron, which is why they are similar metals in the same group. However, sodium's is in the 3s orbital and potassium's is in the 4s orbital, at a different energy level. When excited by the flame, each electron absorbs energy, jumps to a higher orbital, and releases a photon when it falls back. Because the orbital energy gaps are different for the two atoms, the photons have different energies. Sodium releases yellow-orange light (589 nm) and potassium releases violet light. Every sodium atom has the same configuration, so sodium gives the same colour each time.
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Part 4: Isotopes and Relative Atomic Mass

Chlorine atoms from the ocean near Mombasa do not all have the same mass. Data card: Cl-35 (75%) and Cl-37 (25%). (a) Explain what isotopes are and how the two chlorine forms on the card differ. (b) Calculate the relative atomic mass of chlorine and show a reasonableness check. (c) Would Cl-35 and Cl-37 give different flame colours? Explain your answer using electrons.	(a) Isotopes are atoms of the same element with the same atomic number (the same number of protons and electrons) but different mass numbers because they have different numbers of neutrons. Both chlorine isotopes have 17 protons and 17 electrons. Cl-35 has 18 neutrons and Cl-37 has 20 neutrons. (b) Relative atomic mass = $\Sigma(\text{isotope mass} \times \text{fractional abundance}) = (35 \times 0.75) + (37 \times 0.25) = 26.25 + 9.25 = 35.5$. Check: 35.5 lies between the lightest isotope (35) and the heaviest isotope (37), and is closer to 35 because Cl-35 is more abundant. This is why the periodic table gives chlorine as 35.5, not a whole number. (c) No. Both isotopes have 17 electrons in the same arrangement, so their electrons occupy the same energy levels. When heated, they jump to and fall from the same levels and release the same wavelengths of light. Flame colour depends on the electron arrangement, which is set by the atomic number, not by the number of neutrons or the mass number. Flame colour is a fingerprint of atomic number, not atomic mass.
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Part 5: Answering the Driving Question

Write a complete answer to the	Different substances produce different flame colours because their atoms have different electron arrangements. Atomic theory
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driving question: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? Use the cooking hearth in the Rift Valley as your example, and use evidence from across the unit to support your answer.	developed through evidence: Dalton's solid sphere was revised after the gold foil experiment showed a tiny, dense, positive nucleus with electrons in the empty space around it. Bohr then proposed fixed energy levels, and the modern model describes s and p orbitals within them. In the Rift Valley fire, heat is absorbed by the outermost electron of the atom. For sodium this electron is the lone $3s^1$ electron ($1s^2 2s^2 2p^6 3s^1$). The electron jumps to a higher energy level, the excited state. This is unstable, so it falls back to a lower level and releases the energy difference as a photon of light. Sodium's gap corresponds to yellow-orange light of about 589 nm. Potassium, from the banana-peel ash, has its lone outermost electron in the $4s$ orbital ($4s^1$), at a different energy level. Its energy gaps are different, so it releases photons of different energy, which we see as violet. In class, potassium permanganate also gave violet, which fits with potassium being the source of that colour. In the Kisumu laboratory, copper sulphate gave blue-green, because copper has its own energy level spacing. In every substance we tested, the colour was repeatable, because each atom of that element has the same electron arrangement and therefore the same energy gaps. The colour disappears when the substance is removed because no atoms are left to be excited. Isotopes of the same element share the same electron arrangement, so they give the same colour. This is why flame colour is a fingerprint of atomic number and electron arrangement. It reveals that electrons occupy fixed, specific energy levels and orbitals, and not any random positions around the nucleus.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific evidence from several lessons, such as the flame test colours, gold foil results, Bohr diagrams, orbital configurations and the chlorine data, and links each piece to the explanation.	Uses some evidence from the lessons, but the link to the explanation is sometimes missing or general.	Gives little or no evidence, or states facts without any link to the lessons.
Explanation of electron energy transitions	Correctly sequences absorption of energy, jump to the excited state, fall to a lower level and emission of a photon, and explains that the colour depends on the energy gap, which is unique to each element.	Describes the jump and the release of light but is vague about why the colour is specific to the element.	Describes the flame colour without a correct explanation of electron jumps or energy release.
Electron arrangement and configuration	Writes correct configurations for sodium and potassium, applies Aufbau, Pauli and Hund rules, and explains how $3s^1$ versus $4s^1$ leads to different colours.	Writes mostly correct configurations and states the rules, with limited link to the colours.	Configurations are incorrect or missing, and the rules are confused.
Isotopes and relative atomic mass	Calculates 35.5 correctly with a reasonableness check and explains that isotopes share electron arrangement and so the same flame colour.	Calculates correctly but gives an incomplete explanation of the flame colour of isotopes.	Calculation is incorrect or the idea of isotopes is confused.
Scientific communication	Uses precise vocabulary (nucleus, shell, orbital, excited state, photon), a logical order and the Kenyan context throughout.	Uses mostly correct vocabulary and a clear order, with minor errors.	Uses little scientific vocabulary and the ideas are hard to follow.

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